

Information Systems (IS) in the Digital Age

WEEK 01 • LESSON 1

Management of Information Systems (IS) Assessment

Undergraduate · Online · Asynchronous · 120 minutes

What Lesson 1 Covers

Four named parts, plus the assessment method that ties them together

1.1 Information Systems in Global Business Today

What an information system is, what organisations want from one, and the four types by management level.

1.2 Global E-business and Collaboration

Business processes across departments, the four enterprise applications, and how teams actually work.

1.3 Information Systems, Organizations, and Strategy

What organisations are, how systems change them, and the five competitive forces.

1.4 Ethical and Social Issues in Information Systems

The five moral dimensions, the five-step analysis, and the privacy law you will actually meet.

1.5 — Putting it together: the six-lens assessment method

The 120-Minute Running Order

Timings tested against a two-hour block with two short breaks

Minutes	Section	What happens
0 – 8	1.0	Why this course is called **Assessment
8 – 38	1.1	Information Systems in Global Business Today · theory 18 · example 10 · check 2
38 – 42	—	Short break
42 – 70	1.2	Global E-business and Collaboration · theory 16 · example 10 · check 2
70 – 96	1.3	Information Systems, Organizations, and Strategy · theory 15 · example 9 · check 2
96 – 100	—	Short break
100 – 116	1.4	Ethical and Social Issues in Information Systems · theory 10 · example 5 · check 1
116 – 120	1.5	The six-lens assessment method and the Week 02 brief

If you run short of time, cut theory — never cut the examples

Students remember the bike shop. They do not remember the definition of a sociotechnical system. The five worked examples are the load-bearing parts of this lesson.

Learning Outcomes

By the end of this lesson you will be able to do all ten of these

- 1 State**
what an information system is — four activities, three dimensions — without using the word 'computer'
- 2 Distinguish**
data from information and both from knowledge, using your own example
- 3 Name**
the six strategic business objectives and match a real system to two of them
- 4 Identify**
the four system types by management level and which question each answers
- 5 Describe**
a business process crossing departments, and why the crossings lose information
- 6 Explain**
what an enterprise application is and the problem it was invented to solve
- 7 Apply**
the five competitive forces to a small business and recommend one system
- 8 List**
the five moral dimensions and apply the five-step ethical analysis
- 9 Conduct**
a six-lens assessment with a named measurement behind every claim
- 10 Recognise**
the difference between an assessment and an opinion

Why This Course Is Called "Assessment"

An introduction asks you to describe. An assessment asks you to judge — and defend the judgement.

STUDENT A — this is an opinion

"The booking system is quite good. It is easy to use and most students seem to like it. It has improved the library."

Nothing in it can be argued with, because there is nothing in it to argue about. No sentence names anything a sceptical reader could go and check.

STUDENT B — this is an assessment

"Room occupancy rose from 41% to 68% over the same eight weeks before and after launch. Desk staff time on booking disputes fell from 6.5 h/week to 1.0. The claim that students prefer it has never been measured, so it should be removed until it is."

Every sentence names something checkable. One sentence openly admits what is not known.

The property that separates them is FALSIFIABILITY

A claim is falsifiable when it is stated precisely enough that evidence could show it to be wrong.

Student B's paragraph is not longer and not more technical. It is more falsifiable. That single property is the difference between professional judgement and personal impression — and producing it is what you will practise all term.

1.1

Information Systems in Global Business Today

What an information system actually is, what organisations want from one, and why the same software produces different results in two different firms.

MINUTES 8 – 38 · Theory 18 · Example and code 10 · Self-check 2

Data, Information and Knowledge Are Three Different Things

Ordinary speech treats them as synonyms. Systems are built differently for each.

DATA — raw recorded facts

2026-09-14 · JOB-1003 · Marie-Claude
T. · 2.75 · 64.00

No interpretation attached. On their own these five values answer no question. You cannot tell whether the shop had a good day.

INFORMATION — organised to answer a question

"Brake overhauls averaged 2.75 hours in September, against a 2.0-hour estimate."

Many rows grouped and compared against a plan. Different question asked of the same data would have produced different information.

KNOWLEDGE — why it looks that way

"They overrun because the estimate was written for rim brakes, and most bikes now have disc brakes, which take longer to bleed."

No database holds that sentence. A mechanic does. This is why a KMS is the hardest enterprise application to make work.

Why this matters for assessment

When you are told "the system provides good information", the first question is: **which question does it answer, and who asked it?**

A system producing beautiful reports nobody requested is producing data, not information.

What an Information System Actually Is

THE DEFINITION THIS COURSE USES

A set of related components that collect raw facts, transform them, and deliver a result to people who use it to act — together with the feedback that corrects the collection next time.

Notice what is missing: the word "computer". This is deliberate and it is not a trick.

A paper system is still a system

A hospital ward with a paper drug chart, a nurse who reads it, a doctor who signs it, and a review meeting that changes the chart's design is a fully functioning information system.

It has all four activities. It has no computer.

A computer system may not be one

A till that records sales is not yet an information system.

A till that records sales, produces a report showing Tuesdays are dead, and causes the owner to change Tuesday opening hours — that is one. The output changed the input.

The Four Activities — and How Each One Fails

Activity	What happens	Failure mode when it is weak
Input	Raw facts are captured from inside or outside the organisation	Wrong or missing data enters — everything downstream inherits the error
Processing	Facts are sorted, calculated, compared, validated, combined with stored records	An out-of-date rule is applied: the arithmetic is right and the answer is wrong
Output	The result reaches a person in a form they can act on	The answer is correct but arrives too late, in the wrong format, or to the wrong person
Feedback	The result is examined and used to change the input next time	The system runs for years producing the same flawed report because nobody closes the loop

FEEDBACK is the activity beginners leave out — and the one that makes it a system

Two further layers wrap around all four:

- **The organisation** — departments, job titles, rules and habits. These constrain the system far more tightly than the technology does.
- **The environment** — customers, suppliers, competitors, regulators, shareholders. They supply much of the input, consume much of the output, and change without asking permission.

The Three Dimensions — All Present, All The Time

When a system fails, the cause is usually in the dimension nobody was watching

ORGANISATION

People, division of labour, reporting structure, written procedures, unwritten habits.

Includes **business processes** — the sequences by which work actually gets done, which are often different from the sequences written in the procedure manual.

MANAGEMENT

Decisions about what the system is for, who may use it, what happens when it disagrees with a person, and how much money it is worth.

Managers do not merely operate systems. They design the organisation around them — and they are held responsible when that design is wrong.

TECHNOLOGY

Hardware, software, data storage, networks.

The dimension students expect to matter most — and in professional practice, the one that causes the fewest failures.

Technology fails loudly and gets fixed.
Organisational problems fail quietly and persist.

The marking consequence

"It is a PostgreSQL database behind a React front end" describes one dimension out of three. That earns very few marks in this course — not because it is wrong, but because it is a third of an answer.

The Six Strategic Business Objectives (1/2)

Organisations buy systems to achieve one or more of six things. This is the vocabulary of Lens 1.

1

Operational excellence

Do the same work with less waste, less time or fewer errors.

EXAMPLE: A repair shop cuts turnaround from 3 days to 1 by tracking parts in stock.

2

New products, services and business models

Sell something the firm could not previously sell, or sell it a new way.

EXAMPLE: A cinema adds subscription streaming because it now holds the customer data.

3

Customer and supplier intimacy

Know each customer and supplier individually and treat them accordingly.

EXAMPLE: A grocer notices a customer stopped buying and sends a targeted offer.

The Six Strategic Business Objectives (2/2)

4

Improved decision making

Replace a guess with a measurement that arrives in time to act on.

EXAMPLE: A bakery sees hourly sales and stops baking what never sells after 15:00.

5

Competitive advantage

Do something rivals cannot easily copy — and keep doing it.

EXAMPLE: A shipping firm's tracking becomes the reason customers choose it over cheaper rivals.

6

Survival

Adopt it because a law, a regulator or an industry norm leaves no alternative.

EXAMPLE: A clinic adopts electronic records because paper no longer satisfies the regulator.

Assessment lens

A project claiming all six objectives is almost always a project whose sponsor has not decided what it is for. **Claiming two objectives precisely scores higher than claiming six vaguely.**

Complementary Assets — Why Identical Systems Give Different Results

Two rivals buy the same software the same month. One firm's productivity rises. The other's does not move.

ORGANISATIONAL ASSETS

A supportive culture

An efficient business process to run the system inside

Decentralised authority, so decisions can be made where the information arrives

MANAGERIAL ASSETS

Senior management actually backing the project

Incentives that reward using the system rather than working around it

Training programmes

A team that shares knowledge

SOCIAL ASSETS

Internet and telecommunications infrastructure

A labour market with the relevant skills

Technology standards

Laws that make the arrangement enforceable

The practical consequence

The software is rarely the expensive part, and it is almost never the part that fails.

When you assess a proposed system, ask what complementary assets it assumes — and whether the organisation actually has them.

Four System Types, Four Kinds of Question

Who asks the question decides which system answers it

Type	Full name	Used by	The question it answers
TPS	Transaction Processing System	Front-line staff and machines	"Record this sale, this shift, this shipment."
MIS	Management Information System	Middle managers	"How did we do last month against plan?"
DSS	Decision Support System	Analysts and specialists	"What happens if we change this?"
ESS	Executive Support System	Senior management	"Where is this organisation going?"

Data flows UPWARD

MIS is fed by TPS. DSS and ESS are fed by both.
A till operator who rings everything through as "miscellaneous" because it is faster makes every layer above report confident, well-formatted, wrong answers.

MIS means two things

Narrowly: the reporting system in the table above.
Broadly: the name of this entire academic field.
Context tells you which. When writing, say which you mean.

These are not products

No vendor sells "a DSS".
They classify what a system is FOR. One piece of software often does several of these jobs at once.

Two Ways of Studying This Field — You Need Both

THE TECHNICAL APPROACH

Treats systems as formal objects to be optimised. Draws on computer science, management science and operations research.

Asks: is the algorithm correct? Is the model optimal? Is the capacity sufficient?

THE BEHAVIOURAL APPROACH

Treats systems as things that change how people behave. Draws on sociology, psychology and economics.

Asks: who gains power? Who resists? Whose job changes? What will people actually do, rather than what should they do?

THE SOCIOTECHNICAL VIEW — the one this course uses

Neither approach alone predicts what happens. Technology and organisation adjust to each other, and the best outcome usually requires changing **both** rather than optimising either.

The practical rule: when a system fails, look for the human decision that made the technical failure possible. The database did not choose to store MAC addresses. Someone did.

► REAL-LIFE EXAMPLE 1 — Atwater Bike Repair

TASK: run one working day as four separate activities · CONNECTS TO: section 1.1.2, the four-activity definition

A two-person shop on a busy Montréal street. Opens 08:30, closes 17:00. Labour \$62.00/hour. Combined GST + QST 14.975%. Until now: a paper diary and a calculator.

Job	Customer	Service	Accepted	Hours	Parts
JOB-1001	Amélie R.	Tune-up	08:30	1.25	\$18.00
JOB-1002	Dev P.	Flat repair	09:10	0.50	\$12.50
JOB-1003	Marie-Claude T.	Brake overhaul	09:40	2.75	\$64.00
JOB-1004	Sam O.	Wheel true	10:35	0.75	\$0.00
JOB-1005	Priya N.	Full service	14:30	4.00	\$96.75

Look at JOB-1005

Accepted 14:30. Takes four hours. **Finishes 18:30** — ninety minutes after closing.

Somebody stayed late.

On paper, nobody notices this until it has happened eleven times.

The problem is not that the paper diary loses money directly

It is that the diary has **no feedback activity**. The owner notices in March that the shop was busy in September — and by March there is nothing to be done about September.

Example 1 — The Code: `ex1\_1\_ipo\_engine.py`

Python · runs offline · no libraries · every line carries a comment above it

```
# STAGE 1 - INPUT: capture raw facts from the real world
def stage_1_input() -> List[RepairJob]:
    # Build the list of jobs exactly as they arrived at the counter.
    jobs = [ RepairJob("JOB-1001", "Amelie R.", "Tune-up", 1.25, 18.00,
day_start), ... ]
    return jobs

# STAGE 2 - PROCESSING: turn raw facts into something meaningful
LABOUR_RATE_PER_HOUR = 62.00
COMBINED_TAX_RATE    = 0.14975
    # Multiply hours by the rate to get the labour charge for this job.
    labour_charge = job.hours_spent * LABOUR_RATE_PER_HOUR
    # Write the calculated completion time back onto the original record.
    job.completed_at = finished

# STAGE 3 - OUTPUT: present the result to a person who will act on it
    # Add a marker beside jobs that finished after closing time.
    flag = " *" if row["overran"] else "  "

# STAGE 4 - FEEDBACK: send a correction back into tomorrow's input
print("  Change      : stop accepting jobs longer than 3.0 h after 11:00.")
print("  Applied at : STAGE 1 (INPUT) - the counter booking rule.")
```

Run it: python3 ex1_1_ipo_engine.py · Python 3.8+ · no installation · works with no internet connection

Four activities, four functions

Each stage is a separate, clearly named function. You can see exactly where one activity ends and the next begins.

Delete stage 4 and try it

The program still runs. It still prints money.

Ask yourself what the shop has now lost. That is precisely the failure described in section 1.1.2.

Example 1 — What the Program Produced

The output a person reads, and the correction that flows back into tomorrow's input

```
ATWATER BIKE REPAIR - DAILY JOB SHEET - 14 September 2026
=====
JOB      CUSTOMER      SERVICE      LABOUR  PARTS  TOTAL  DONE
JOB-1001  Amelie R.      Tune-up      77.50   18.00  109.80 09:45
JOB-1002  Dev P.         Flat repair   31.00   12.50  50.01 09:40
JOB-1003  Marie-Claude T. Brake overhaul 170.50   64.00  269.62 12:25
JOB-1004  Sam O.         Wheel true    46.50    0.00  53.46 11:20
JOB-1005  Priya N.       Full service  248.00   96.75  396.38 18:30 *
-----
REVENUE (incl. tax)                879.27
  of which sales tax to remit        114.52
```

FEEDBACK — the fourth activity

```
Finding      : 1 job finished after closing.
Job(s)       : JOB-1005
Change       : stop accepting jobs > 3.0 h
               after 11:00.
Applied at   : STAGE 1 (INPUT)
```

The output has just changed the input. That is what makes it a system.

\$879.27

Day revenue including tax

\$114.52

Sales tax to remit — money that is not the shop's

1 of 5

Jobs that overran closing time, caught on day one

Lesson 1.1 — Self-Check and Video

SELF-CHECK — answer these before moving on

1. Give one example each of data, information and knowledge from your own week. Explain what makes each belong to its category.
2. A supermarket till records every scanned item. Is that an information system? Justify using the four activities.
3. Which of the six business objectives does a payroll system serve? Name at least two and say why.
4. A firm buys the same inventory software as its competitor and gets worse results. Name three complementary assets that could explain it.
5. Why does bad data in a TPS matter more than bad data in an ESS?

Worked answers are at the end of the written pack.



VIDEO FOR LESSON 1.1

Primary link

youtube.com/watch?v=a5_ORxT4eyA

Permanent fallback (a search page — cannot break)

youtube.com/results?search_query=information+systems+in+global+business+today+chapter+1

Watch for: the presenter's treatment of the organisation / management / technology triangle. Note any point where it differs from ours — that difference is a good discussion question.

Scan every link at [virustotal.com](https://www.virustotal.com) before opening.

1.2

Global E-business and Collaboration

Why a business is best understood as a set of processes that cross departments, and why almost the entire enterprise software industry exists because of that.

MINUTES 42 – 70 · Theory 16 · Example and code 10 · Self-check 2

A Business Is a Set of Business Processes

A specific sequence of steps, performed by specific people, producing something someone needs

Organisations describe themselves by DEPARTMENTS. Customers experience PROCESSES that run across those departments.



That whole sequence is called ORDER-TO-CASH. It exists in essentially every commercial organisation on earth.

Other processes you will meet

Procure-to-pay — deciding to buy through to paying the supplier

Hire-to-retire — recruiting an employee through to their departure

Record-to-report — recording transactions through to publishing statements

THE CRITICAL POINT

The places where a process crosses from one department to another are the places where information gets lost, delayed, duplicated or contradicted.

Almost the entire enterprise software industry exists to address that single fact.

Functional Systems and Information Silos

Each function bought its own system, in isolation, at a different time, from a different vendor

Function	What its system does	Typical records it holds
Sales and marketing	Identify customers, take orders, track campaigns	Customer · order · quote · campaign
Manufacturing and production	Plan and schedule the making of things	Bill of materials · work order · machine schedule
Finance and accounting	Record money in and out, produce statements	Invoice · payment · ledger entry · tax record
Human resources	Hire, pay, develop and track employees	Employee · contract · payroll run · training record

INFORMATION SILOS

Each of those systems works perfectly on its own. The problem is what happens **between** them.

If Sales holds one customer list and Finance holds another, the two lists will disagree within weeks — and no amount of care by either department will stop it. There is no shared record for either of them to be right about.

This is not caused by bad staff or bad software. It is caused by each function optimising its own work without a shared record.

The Four Enterprise Applications (1/2)

All four share one idea — a single shared store of data every function reads and writes

ERP — Enterprise Resource Planning

One integrated system, on one database, covering most core functions at once: finance, human resources, manufacturing, sales and distribution.

When Sales records an order, Finance and Supply see it in the same instant — because there is only one record and all three are looking at it.

Examples: SAP · Oracle Fusion · Microsoft Dynamics

Open alternatives: Odoo · ERPNext

THE TRADE-OFF — state this in any assessment: an ERP system removes the walls between departments, and it also removes their independence. Every department must accept a common definition of "customer", "order" and "revenue". That negotiation is usually harder than the installation.

SCM — Supply Chain Management

Coordinates suppliers, production and distribution so that material arrives when it is needed rather than when it happens to be ordered.

SCM extends the information system **beyond the walls of the firm.**

The consequence, which is the part worth remembering: the firm now depends on data quality it does not control. A supplier who updates their stock figures twice a week has just become your data quality problem.

Assessment question to ask: whose data feeds this, and what happens to your decisions when their data is late?

The Four Enterprise Applications (2/2)

CRM — Customer Relationship Management

Stores every interaction with a customer so that any employee can see the full history.

This converts a customer relationship from something living in one salesperson's memory into an **organisational asset that survives that salesperson leaving**.

Examples: Salesforce · HubSpot

Open alternatives: SuiteCRM · EspoCRM

KMS — Knowledge Management System

Captures expertise and documents so that knowledge survives an individual employee's departure.

The hardest of the four to make succeed, for a simple reason: it depends on busy people writing down what they know, and writing things down takes time nobody is paid for.

How to assess one: ask who is given time to contribute to it. If the answer is nobody, the system will be empty within a year.

Every tool named in this course is an example, not a recommendation

Each category has commercial products, open-source products, and the option of building your own. Naming a product is not an answer; naming a category and justifying a choice within it is.

Intranets, Extranets and the Three "E-" Words

INTRANET vs EXTRANET

Intranet — a private network using ordinary web technology, restricted to people inside the organisation. Handbooks, directories, expense forms.

Extranet — the same thing, deliberately extended to specific outsiders. A supplier portal where a vendor logs in to see purchase orders is an extranet.

The interesting question is not technical

What have you now made a business partner dependent on, and what happens to them if you switch it off?

An extranet creates a relationship that is harder to leave. In Lesson 1.3 this is called a **switching cost** — and creating them deliberately is a recognised strategy.

E-BUSINESS

Digital technology running **any** organisational process, internal or external. A factory scheduling system is e-business. No selling involved.

E-COMMERCE

The subset involving buying and selling over a network. **B2B · B2C · C2C**. By transaction value B2B is by far the largest — which surprises most students.

E-GOVERNMENT

The same technology used by public bodies to deliver services to citizens and businesses.

Two properties that explain much of what follows

Marginal cost — the cost of one more unit — is near zero for digital goods, so a software company serves its millionth customer at almost no extra cost. And a network's usefulness grows as more people join, which is why the second messaging app on a campus struggles even when it is better than the first.

Collaboration — and the Time / Space Matrix

Working with others toward a shared and explicit goal. Now the activity most systems are actually used for.

Why it has grown in importance

Work changed from **routine** to **interpretive**. Routine work can be assigned to one person; interpretive work requires several people arguing productively.

Organisations **flattened** — decisions that once travelled up a hierarchy get made by the team holding the information.

Firms are **geographically distributed**. A team spanning Montréal, Lisbon and Bangalore has no shared corridor.

Innovation is now understood as a group process, not an individual one.

	SAME TIME	DIFFERENT TIME
SAME PLACE	A meeting room with a whiteboard	A shared physical board; a shift handover book
DIFFERENT PLACE	Video call; live shared document; screen sharing	Email; ticket system; wiki; threaded chat

Different time + different place — the hardest quadrant, and the only one that scales across time zones

THE MEASUREMENT PROBLEM

Collaboration tools are notoriously hard to assess because their benefits are indirect. Measure a **process outcome**, not the tool.

Not "how many messages were sent" — that number always goes up. Instead: **how long does a decision now take, from question asked to answer accepted?**

Who Actually Runs the Information Systems Function

Assessments must name who is accountable — and "the IT department" is not a name

Role	Full title	Accountable for
CIO	Chief Information Officer	The whole IS function and its alignment with business goals
CSO	Chief Security Officer	Protecting information and physical assets; security policy
CPO	Chief Privacy Officer	Lawful and fair handling of personal data
CKO	Chief Knowledge Officer	Capturing and sharing organisational knowledge
CDO	Chief Data Officer	How data is collected, governed and used across the firm

Below the executives

Systems analysts — translate business problems into system requirements. The single most important translation in this field.

Programmers · network specialists · database administrators

AND THE LARGEST GROUP OF ALL: END USERS

Everyone outside the function who uses a system to do their job.

End users are **not passive recipients**. In most organisations they are the largest single source of both requirements and unofficial workarounds — and the workarounds tell you more about the system than the requirements do.

► REAL-LIFE EXAMPLE 2 — Rue Sherbrooke Café

TASK: follow one order across four departments · CONNECTS TO: sections 1.2.1 and 1.2.3

About 300 customers a day. Four people on shift. One owner who does the books on Sunday evening. Until last year it ran on four separate lists — all describing the same business, all disagreeing by Friday.

Counter

paper order pad

Kitchen

whiteboard

Card terminal

its own daily total

Stockroom

a clipboard, updated when remembered

The consequences were not dramatic — they were routine and expensive

Sunday: the card total does not match the order pad. Ninety minutes finding out why.

Thursday: milk runs out because the clipboard has not been updated since Tuesday.

A catering order is prepared twice because the closing shift wiped the whiteboard.

None of these are technology failures

Every one of them is a **handoff failure** — information not surviving the crossing from one function to the next.

The café now runs one system where all four functions read and write the same records.

Figure 3 in the pack is a SWIMLANE DIAGRAM

Each horizontal band is one department; the arrows show where work crosses from one band to the next. **Every one of those arrows is a place where information can be lost.**

Example 2 — The Code: `ex1\_2\_order\_to\_cash\_console.html`

Self-contained HTML · double-click to open · no scripts, no fonts, no network requests of any kind

```
/* Define the function that runs one complete order through all four lanes. */
function placeOrder(description, price, consumes){
  /* ---- LANE 1: SALES ---- */
  logLine('sales', 'Order #' + orderNo + ' accepted: ' + description);
  /* ---- LANE 3: FINANCE ---- */
  const tax    = price * TAX_RATE;
  const grand = price + tax;
  /* ---- LANE 4: SUPPLY ---- */
  /* Only deduct stock if the shared database is working. */
  if(!sharedBroken){
    stock.milk -= consumes.milk;  stock.beans -= consumes.beans;
    logLine('sup', 'Consumption received; stock updated');
  } else {
    /* When sharing is broken the message never arrives, so warn loudly. */
    logLine('warn', 'Consumption message LOST - Supply never heard about this
order');
  }
}
```

What you can do with it

Place an order and watch it cross all four lanes, with a timestamped audit log of every handoff and live stock levels that trigger purchase orders.

☒ Break the shared database

Tick the box. Sales, Operations and Finance keep working perfectly. The screen looks entirely normal. The stock figures quietly become wrong.

That silence is why silo failures are expensive.

WHY HTML AND NOT PYTHON HERE: this example is about four things happening at once and needing to be seen side by side. A terminal prints one line after another; a web page shows four panels updating together. The language follows the teaching need, not the other way round.

Lesson 1.2 — Self-Check and Video

SELF-CHECK

6. Name the four steps of order-to-cash at a business you used this week, and which department performs each.
7. What problem was the ERP category invented to solve? One sentence, without using the word "integrated".
8. Give one advantage and one genuine disadvantage of putting every department on a single ERP system.
9. Which quadrant of the collaboration matrix does this course occupy, and why is it the hardest?
10. Why is "number of messages sent" a poor measurement of a collaboration tool? Propose a better one.



VIDEO FOR LESSON 1.2

Primary link

youtube.com/watch?v=X1OhamT8s3k

Permanent fallback

youtube.com/results?search_query=business+process+management+explained+beginners

Watch for: how the presenter draws a process that crosses departments. Compare their diagram against Figure 3 and identify which handoffs their version leaves out.

Scan every link at [virustotal.com](https://www.virustotal.com) before opening.

1.3

Information Systems, Organizations, and Strategy

What an organisation actually is, how systems change its shape, and the framework for working out where a firm's money is going before deciding what to do about it.

MINUTES 70 – 96 · Theory 15 · Example and code 9 · Self-check 2

Two Definitions of an Organisation — You Need Both

THE TECHNICAL DEFINITION

A stable, formal structure that takes resources from its environment and processes them into outputs. Capital and labour in; products and services out.

Precise. Explains why organisations can be redesigned like machinery: change the inputs or the processing rules and the outputs change.

THE BEHAVIOURAL DEFINITION

A collection of rights, privileges, obligations and responsibilities balanced over a long period through conflict and negotiation.

Less tidy. Considerably more useful when predicting what will actually happen.

Why you need both

The technical definition tells you what a system **could** change. The behavioural definition tells you what it **will be permitted** to change.

A new system that reassigns who approves a purchase has not merely changed a workflow. It has taken a privilege from one person and given it to another.

Whether the technical design is elegant is largely irrelevant to whether it survives.

Features Every Organisation Has

1

ROUTINES AND BUSINESS PROCESSES

Rules developed over time for handling expected situations. An organisation is, in practical terms, a large bundle of routines — which is why it is efficient at what it does repeatedly and clumsy at anything new. The routines that make the first possible make the second difficult.

2

ORGANISATIONAL POLITICS

Different positions produce genuinely different views, because different positions are held accountable for different things. A finance director who resists a system that reduces financial control is doing their job. This is a design problem, not a character problem.

3

ORGANISATIONAL CULTURE

Assumptions about what the organisation is for that nobody articulates because everyone shares them. A powerful stabiliser and a powerful brake: technology that challenges a cultural assumption meets resistance from people who cannot say why they object.

4

ENVIRONMENTS

Organisations and environments shape each other. Environments generally change faster, which is the main reason organisations fail. Note that organisations shape environments too — they lobby, they set standards, they build constraining relationships.

How Systems Change Organisations — Two Economic Explanations

TRANSACTION COST THEORY

Firms exist because buying things in a market costs money — finding suppliers, negotiating, writing contracts, checking that what arrived is what was promised. Those are **transaction costs**.

When they are high it is cheaper to do the work inside, so firms grow. Information systems lower them, making buying from outside cheaper relative to doing it inside.

Observable consequence: firms stay smaller while handling more business, and outsource more of what they once did internally.

AGENCY THEORY

An owner cannot do everything, so they hire agents. Agents have their own interests, so the owner must spend money supervising them — **agency costs**.

As a firm grows, agency costs grow faster than the firm, which historically limited how large a firm could get. Systems reduce the cost of supervision.

Observable consequence: flatter organisations with fewer middle-management layers.

Both predict the same outcome, differently

Smaller, flatter organisations handling more work — for different reasons.

Being able to name which mechanism is operating in a specific case is worth marks.

RESISTANCE TO CHANGE

A system changes four things at once — **task, technology, people, structure** — and all four are interlocked.

When a system "fails", ask which of the four changed and which did not. A rejected system is not fixed with more software.

The Five Competitive Forces (1/2)

Michael E. Porter, Harvard Business School, 1979 · isc.hbs.edu · cite Porter directly in written work

RIVALRY AMONG EXISTING COMPETITORS

1

How hard the firms already in the market compete. High rivalry pushes prices toward cost.

IS RESPONSE

Systems that make your offer non-comparable — loyalty programmes holding purchase history, pricing that adjusts by segment.

THREAT OF NEW MARKET ENTRANTS

2

How easily a new firm can start competing. Where start-up costs are low, profit attracts imitators quickly.

IS RESPONSE

Systems that raise SWITCHING COSTS — stored value, saved preferences, accumulated history. All make leaving expensive.

THREAT OF SUBSTITUTE PRODUCTS OR SERVICES

3

Whether customers can meet the same need a different way. A bubble tea shop competes with canned drinks at the corner store.

IS RESPONSE

Systems that shorten the time from noticing a demand shift to launching a product that meets it.

The Five Competitive Forces (2/2)

BARGAINING POWER OF CUSTOMERS

4

How much leverage buyers have to push prices down. Power rises when switching is easy and comparison is easy — and comparison has become nearly free.

IS RESPONSE

CRM segmentation, personalised offers, and service that makes price the second question rather than the first.

BARGAINING POWER OF SUPPLIERS

5

How much leverage sellers have to push prices up. Power rises when there are few suppliers.

IS RESPONSE

Supplier portals with price history and multi-source quoting, which turn a single relationship into a comparable market.

HOW TO READ A FIVE-FORCES ANALYSIS

A **strong** force means value is leaving your firm through that channel. A **weak** force means value stays.

The framework does not tell you what to do. It tells you **where the money is going** — which is a prerequisite for deciding what to do.

Four Generic Strategies — and the Honest Caveat

LOW-COST LEADERSHIP

Produce at the lowest cost in the industry, so you survive a price war others cannot.

SYSTEMS: Inventory replenishment · supplier portals forcing competitive quoting · process automation removing handling steps

PRODUCT DIFFERENTIATION

Offer something rivals do not have, so price is not the deciding factor.

SYSTEMS: Mass customisation · product data analysis shortening idea-to-launch time

FOCUS ON A MARKET NICHE

Serve a narrow segment better than any general competitor can.

SYSTEMS: Fine-grained customer analysis to identify a segment worth owning, then to serve it specifically

CUSTOMER AND SUPPLIER INTIMACY

Build relationships so close that leaving becomes expensive for the other party.

SYSTEMS: CRM · extranets · any system accumulating history the counterpart cannot take with them

THE CAVEAT MOST TEXTBOOKS SKIP

Buying a system does not produce a strategy. Every competitor can buy the same software from the same vendor within a quarter. Advantage comes from the complementary assets — process redesign, training, incentive changes — which are far harder to copy than a licence.

Value Chain, Value Web, and Three Further Ideas

THE VALUE CHAIN MODEL

Primary activities — inbound logistics · operations · outbound logistics · sales and marketing · service

Support activities — firm infrastructure · human resources · technology development · procurement

Its analytical use is specific: instead of "where should we apply technology?" (answer: everywhere), it asks "**which single activity, if improved, would most change what the customer experiences or what the firm spends?**" That is answerable.

THE VALUE WEB

Modern products are produced by **networks of firms** whose systems are connected — suppliers, distributors, logistics providers, partners — coordinated through shared systems rather than through ownership.

The strategic consequence: a firm's competitive position now depends partly on the quality of systems it does not own and cannot directly control.

Benchmarking — comparing one activity against an industry standard — is how you tell whether an activity is genuinely weak or merely unfamiliar.

NETWORK ECONOMICS

Each additional participant makes the network more valuable to every existing one, at near-zero cost. This is why platform markets tend toward a few large winners rather than many medium ones.

BUSINESS ECOSYSTEMS

Loosely coupled interdependent firms around a KEYSTONE firm supplying a platform, plus NICHE firms building on it. Assessing a niche firm means assessing its dependence — a risk on no balance sheet.

DISRUPTIVE TECHNOLOGY

Arrives worse than what exists, serving a market incumbents do not want, and improves until it displaces them. Every incumbent decision to ignore it was rational at the time. "The managers were stupid" is almost always wrong.

► REAL-LIFE EXAMPLE 3 — Two Bubble Tea Shops, One Street

TASK: score both against all five forces with evidence · CONNECTS TO: sections 1.3.5 and 1.3.6

BOBA BOULEVARD — open six years

Cash and card. Paper loyalty card that gets stamped and frequently lost. **One** tapioca importer, who raised prices 18% last quarter with no notice and no negotiation. The owner is well liked and works extremely hard.

PEARL & CO. — open two years

Mobile app storing value, remembering previous orders, issuing personalised offers. **Three** suppliers, each required to quote before ordering. The owner reviews sales figures every Monday for twenty minutes.

Force	Boba	Pearl	Why the difference
Rivalry among existing competitors	4	3	41% of Pearl's sales come from app regulars who do not price-shop
Threat of new market entrants	5	2	Pearl's customers hold an average \$14 of stored credit; leaving costs them
Threat of substitute products	3	3	Canned tea at the dépanneur affects both equally — the app does not fix this
Bargaining power of customers	4	2	Personalised offers mean fewer Pearl customers decide on price alone
Bargaining power of suppliers	5	2	Pearl requires three quotes; the last price rise was declined
TOTAL PRESSURE	21/25	12/25	

Notice the row where the scores are IDENTICAL

An honest assessment names the thing the intervention did **not** fix. Reports showing improvement in every row are usually reports where the scoring was done after the conclusion.

Example 3 — The Code: `ex1\_3\_five\_forces\_scorecard.py`

Python · scores, evidence, verdict, priority force, and the system to build first

```
# Map each force to the system that most directly reduces it, and to the
# generic strategy that system serves. This mapping IS the lesson.
IS_RESPONSE = {
    "Threat of new market entrants": (
        "Mobile app with stored value and rewards (raises switching cost)",
        "Customer and supplier intimacy"),
    "Bargaining power of suppliers": (
        "Supplier portal with price history and multi-source quotes",
        "Low-cost leadership"),
}

# A score with no evidence is worth nothing.
evidence={ "Bargaining power of suppliers":
    "Single tapioca importer raised prices 18 percent with no notice." }

# Find the single force pressing hardest on a firm.
def weakest_position(firm): return max(firm.scores, key=firm.scores.get)
```

Evidence is not optional in the DATA STRUCTURE

It sits beside every score. A design choice that enforces a professional habit at the level of the code itself.

Written out so you can DISAGREE

The mapping is deliberately explicit. Disagreeing with it, in writing, with evidence — that is the assessment task.

TRY THIS (five minutes)

Change Boba Boulevard's supplier score from 5 to 2 — they found a second importer. Re-run. The PRIORITY FORCE changes, so a **different system** is now recommended. One operational decision has just redirected the entire technology budget.

Lesson 1.3 — Self-Check and Video

SELF-CHECK

11. Take a shop or service you use regularly. Score all five forces 1–5 with one sentence of evidence each. Which force is strongest?
12. Explain — without using "integrated" or "synergy" — why lowering transaction costs allows a firm to stay smaller.
13. A firm installs the identical inventory system as its rival and gains no advantage. Explain why, using section 1.1.5.
14. Which generic strategy is a public college pursuing when it improves room booking? Defend your answer, including its weaknesses.
15. Give a switching cost you personally bear right now, and estimate what leaving would cost you.



VIDEO FOR LESSON 1.3 — the best one

The Explainer: The 5 Forces That Make Companies Successful — Harvard Business Review, official channel

youtube.com/watch?v=XCWHSeDU-zk

Published by the journal that first published the framework. Under five minutes. **The single most stable link in this course.** If you watch only one video this week, watch this one.

Watch for: it never mentions information systems. Add that layer yourself — for each force it names, state which system category from Lesson 1.2 addresses it.

1.4

Ethical and Social Issues in Information Systems

Why ethical questions in this field arrive as side effects rather than as decisions, and how to turn a question of taste into a question with a findable answer.

MINUTES 100 – 116 · Theory 10 · Example and code 5 · Self-check 1

What "Ethics" Means Here — and Why It Is Not Optional

THE DEFINITION

The set of principles a person uses to choose between courses of action when both are available and they are responsible for the choice. **It is not the same as law.** Something can be legal and wrong; something can be illegal and defensible. Ethics is what you use in the very large space where the law says nothing.

Why that space is so large in this field

Information systems make new actions cheap enough to be worth doing.

Recording where every student sits in a library was always technically possible — someone could have stood in the doorway with a clipboard. It was not worth doing.

Sensors made it cost almost nothing

And something that costs almost nothing gets done **without anybody deciding to do it.**

Nobody at the college held a meeting and resolved to track students. Somebody chose a sensor model, and the tracking came included.

THE ASSESSMENT CONSEQUENCE

An ethical question in an information system is almost never **"should we do this obviously bad thing?"**

It is **"who decided this, when, and did they know they were deciding it?"** — a question with a findable answer, which is why it belongs in an assessment rather than in a debate.

The Five Moral Dimensions

Lens 5 of the assessment framework · each one is a different kind of question

1

INFORMATION RIGHTS AND OBLIGATIONS

What may be collected about a person, what must be disclosed, and what claims they retain over data about themselves. PRIVACY is the claim to be left alone and free from surveillance.

2

PROPERTY RIGHTS AND OBLIGATIONS

Who owns data, derived insights and software. Digital things copy perfectly at no cost, which makes traditional property rules hard to apply and harder to enforce.

3

ACCOUNTABILITY, LIABILITY AND CONTROL

Who answers for a wrong outcome, who bears the cost, and who has the power to reverse it. This is the dimension that automation most reliably erodes.

4

SYSTEM QUALITY

What standard of accuracy and reliability is good enough given the consequences. No system is perfect; the question is what error rate is acceptable for the harm it can do.

5

QUALITY OF LIFE

Which values, habits and institutions the system changes as a side effect. The broadest and least measurable — and where students most often stop thinking because it feels vague.

The Technology Trends That Raise the Stakes

Each one lowers the cost of an action that was previously too expensive to bother with

Trend	What it makes cheap	The ethical consequence
Rising computing power	Analysing everything rather than sampling	Nothing needs to be discarded, so nothing is
Falling data storage cost	Keeping records forever	"Delete when no longer needed" stops being the default
Advances in data analysis	Finding patterns across separate datasets	PROFILING — a detailed picture built from many small, individually harmless records
Networking advances	Moving data anywhere instantly	Data crosses legal jurisdictions without anybody noticing
Mobile devices	Continuous location and device identification	Tracking becomes the default rather than a decision

PROFILING

Combining data from several sources to build a detailed picture of an individual.

The point students miss: each contributing record can be entirely innocuous. Room bookings are not sensitive. Wi-Fi logs are not sensitive. Door sensor events are not sensitive. Combined, they describe where a named person was, for how long, and with whom — which is sensitive, and which nobody ever decided to collect.

NON-OBVIOUS RELATIONSHIP AWARENESS

Techniques that find connections between people that neither person declared.

Legitimately valuable in fraud detection. Legitimately dangerous everywhere else.

And it is the same technique used for both.

Responsibility, Accountability, Liability — Three Different Words

RESPONSIBILITY

You accept the potential costs and duties of a decision.

Something a **person** takes on.

ACCOUNTABILITY

A mechanism exists to determine who took the action.

A property of a **system**, not of a person. No mechanism, no accountability — regardless of who feels responsible.

LIABILITY

A legal framework permits recovering damages from the party at fault.

Exists **only** where law provides it.

THE ACCOUNTABILITY GAP

A system takes an action, the action harms someone, and no mechanism exists to determine which human decision produced it.

"The algorithm did it" is a description of an accountability gap, not an explanation of one.

THE RULE THIS COURSE APPLIES

Every automatic decision must have a named human who can reverse it. Where a system cancels a booking, refuses an application or flags an account with no human able to undo it, the design has a defect regardless of how accurate it is.

The Five-Step Ethical Analysis

Work through these in order and write down each one. The writing is what makes the reasoning checkable later.

1

Identify and describe the facts clearly

Who did what, to whom, where, when, how. Most apparent ethical disputes turn out to be factual disputes wearing an ethical costume — and this step resolves them.

2

Define the conflict and identify the higher-order values

Ethical questions are rarely good against bad. They are usually one legitimate value against another — safety against privacy, efficiency against fairness. Name both.

3

Identify the stakeholders

Everyone with an interest in the outcome. The step most commonly skipped and the one that most changes the answer, because it forces you to name somebody who is not in the room.

4

Identify the options you can reasonably take

There are almost never two. There are usually four or five — and the good one is usually the third.

5

Identify the potential consequences of each option

For each stakeholder, not only for the organisation.

Sheet '06 Ethics Log' of the workbook is a decision log with these five steps as its columns, plus a fully worked example row.

Six Candidate Ethical Principles

None resolves every case. They are tests you apply — and when several agree, the answer is reasonably safe.

THE GOLDEN RULE

Do to others as you would have them do to you.

IN PRACTICE: Put yourself in the position of the person the data is about.

KANT'S CATEGORICAL IMPERATIVE

If an action is not right for everyone to take, it is not right for anyone.

IN PRACTICE: If every college did this, would the result be acceptable?

THE SLIPPERY SLOPE RULE

If an action cannot be repeated indefinitely without harm, it should not be taken at all.

IN PRACTICE: This collection is fine at one field — what stops the tenth?

THE UTILITARIAN PRINCIPLE

Take the action achieving the greatest value for the greatest number.

IN PRACTICE: Rank the outcomes, choose the highest total. Known weakness: it can justify harm to a minority.

THE RISK AVERSION PRINCIPLE

Take the action producing the least harm or lowest potential cost.

IN PRACTICE: Prefer the option whose worst case is survivable.

THE "NO FREE LUNCH" RULE

Assume anything someone else created belongs to them unless specifically declared otherwise.

IN PRACTICE: Absence of a copyright notice is not permission.

HOW TO USE THESE IN WRITTEN WORK

Do not list all six. Choose two that point in **different** directions, apply both, and explain which you followed and why. An analysis where every principle agrees is usually an analysis of an easy case.

Professional Codes and the Privacy Law You Will Actually Meet

PROFESSIONAL CODES OF CONDUCT

ACM — acm.org — Code of Ethics and Professional Conduct, the most cited in computing

IEEE — ieee.org — and IEEE Computer Society, computer.org

CIPS — cips.ca — the Canadian professional body

A code is not law, and breaching one usually carries no legal penalty. Its real value: it gives you something **external** to point at when instructed to do something you should not do. "I would prefer not to" is weak. "This breaches clause 1.6 of the code I am bound by" is much stronger.

FIP — Fair Information Practices

Not a law, but the principles underlying nearly all of them: tell people what you collect and why; collect only what you need for that purpose; let people see and correct their own data; secure it; be accountable for it.

Almost every statute is a specific implementation of those five ideas.

Jurisdiction	Instrument and regulator
Canada — federal	PIPEDA · Office of the Privacy Commissioner (priv.gc.ca)
Québec	Law 25 amendments · Commission d'accès à l'information (cai.gouv.qc.ca)
European Union	GDPR, in force May 2018 · reaches organisations outside the EU
United States	No single federal statute · CCPA is the most influential state law
International	OECD 1980 guidelines · NIST Privacy Framework (free)

TWO MECHANISMS WORTH KNOWING BY NAME

Informed consent — agreement given with knowledge of the facts. A consent buried in twelve thousand words is agreement without knowledge, which is not informed consent regardless of what was clicked.

Opt-in vs opt-out — the default setting determines the outcome for the large majority of people. Which is why the choice of default is an **ethical** decision, not a user-interface decision.

Intellectual Property — Three Protections, Three Purposes

Instrument	What it protects	How long	The catch
Trade secret	Any formula, method or compilation kept confidential	As long as it stays secret	One leak and the protection is gone permanently
Copyright	The **expression** of an idea — code, text, images	Decades beyond the author's life (varies by country)	Protects the expression, never the underlying idea
Patent	An invention: a novel, useful, non-obvious method or device	Roughly 20 years from filing	Requires public disclosure of exactly how it works

THE DISTINCTION THAT MATTERS MOST FOR SOFTWARE

Copyright protects how you wrote it, not what it does.

Someone may lawfully read your program, understand the idea, and implement the same idea in their own code. The idea is not yours. The expression is.

Relevant bodies

WIPO — wipo.int — the UN agency for intellectual property, international position
Canadian Intellectual Property Office — registers Canadian patents, trademarks, copyright

System quality is an ethical question too

Beyond a point, removing the next defect costs more than any benefit. That makes "how good is good enough?" an **ethical** decision, because the answer depends on what the system can do to people.

► REAL-LIFE EXAMPLE 4 — The "StudySpot" Campus Application

TASK: privacy impact self-audit across five moral dimensions · CONNECTS TO: sections 1.4.2 – 1.4.5

StudySpot lets students book study rooms. It replaced a paper sign-up sheet and it works well — desk time on booking disputes fell from about 6.5 hours a week to about 1.

Student number

Room

Time slot

MAC address

▲ look at this one

The booking function works perfectly without it

It was included because the analytics vendor's software collects it **by default**, and nobody switched it off.

MAC = Media Access Control — the unique hardware identifier of a device's network card.

Consider what became possible

A MAC address identifies a **device**, and a device usually identifies a person. Cross-referenced with campus Wi-Fi logs, the booking records now describe where a named individual was, for how long, and — via adjacent bookings — plausibly with whom.

Nobody designed that capability. Nobody approved it.

It arrived as a **default setting in a procurement decision** — exactly the mechanism described in section 1.4.1.

The chosen option: collect a rotating identifier that cannot be linked to a person. It survives Risk Aversion, the Golden Rule and Kant's test simultaneously. **It costs two days of work.**

Example 4 — The Code: `ex1\_4\_privacy\_impact\_audit.html`

Self-contained HTML · twelve weighted questions · nothing you type leaves your browser tab

```
/* The structure of the DATA is the structure of the THEORY. */
const DIMENSIONS = [
  { name: "1. Information rights and obligations",
    items: [
      {q:"Is the MAC address genuinely required to book a room?", w:3},
      {q:"Is the purpose stated in plain language before collection?", w:3},
    ] },
  { name: "3. Accountability and control",
    {q:"Is one named person accountable for data decisions?", w:3},
  } ];

/* Three answers, three risk multipliers. "Partly / unsure" exists because
   it is the honest answer to most real audit questions, and an instrument
   that forbids honesty collects lies. */
const ANSWERS = [ {label:"Yes", risk:0}, {label:"Partly / unsure", risk:0.5},
                   {label:"No", risk:1} ];
```

It refuses to score an incomplete audit

Deliberate. A partial audit producing a confident-looking percentage is worse than no audit — because somebody will quote the percentage.

It ranks dimensions, not just the total

40% overall could mean four dimensions at 10% and one at 100%. The breakdown exists so that difference cannot hide.

TRY THIS (five minutes)

Answer "Yes" to everything except `*"Is the MAC address genuinely required?"`. Look at the resulting score. **One unnecessary data field produces measurable risk all by itself** — which is the practical meaning of the principle of data minimisation.

Lesson 1.4 — Self-Check and Video

SELF-CHECK

16. Explain the difference between responsibility, accountability and liability, with one example of your own for each.
17. Name three separate, individually harmless records held about you that would, combined, describe your daily movements.
18. Apply Kant's Categorical Imperative and the Utilitarian Principle to the MAC address question. Do they agree? Which would you follow?
19. A company's terms run to 14,000 words and a user clicks "I agree". Has informed consent been given? Justify.
20. Give one example of a system that is legal, technically excellent, and still ethically indefensible. Which moral dimension does it fail?



VIDEO FOR LESSON 1.4

Primary link

youtube.com/watch?v=LQeSExn1Dtw

RECOMMENDED OVER ANY VIDEO: the European Data Protection Supervisor (edps.europa.eu) and the Office of the Privacy Commissioner of Canada (priv.gc.ca) both publish short, plain-language explanations. For this topic a regulator's own explanation is more reliable than any video — and both are free.

Watch for: whether the presenter distinguishes **legal** from **ethical**. Many treatments collapse the two, and that collapse is the commonest error in student work on this subject.

1.5

Putting It Together: The Six-Lens Assessment Method

Everything above was preparation for this. The four parts supply the vocabulary; this supplies the method that turns vocabulary into a defensible judgement.

MINUTES 116 – 120 · The method you will use for the rest of the course

The Six Lenses — the Marking Scheme for Every Graded Task

1

PURPOSE

Which of the six business objectives does this system serve?

Lesson 1.1.4

2

PEOPLE

Who inputs, who decides, and who is affected?

Lessons 1.1.3 · 1.2.7

3

PROCESS

Which steps does it change, remove or create?

Lesson 1.2.1

4

STRATEGY

Which competitive force does it weaken?

Lesson 1.3.5

5

ETHICS

Whose data, whose risk, whose accountability? ×2 WEIGHT

Lesson 1.4.2

6

EVIDENCE

What measurement proves each claim is true? ×2 WEIGHT

Lesson 1.0

FOUR RULES THAT GOVERN THE MARKING

1. A lens you skip is a mark you lose. All six, every time.
2. Ethics and Evidence carry DOUBLE weight — those are where assessments fail in practice.
3. A score with no written finding scores ZERO. Not reduced — zero.
4. Score the WEAKEST lens, not the average. Averaging hides what you were asked to find.

► REAL-LIFE EXAMPLE 5 — Full Assessment of "LibreDesk"

TASK: complete six-lens assessment · CONNECTS TO: the synthesis of all four lesson parts

1. Purpose		9/10	Two objectives claimed, both plausible, neither overstated
2. People		8/10	Six roles; the privacy officer explicitly named accountable
3. Process		8/10	Six steps — but step 4 cancels a booking with nobody able to reverse it
4. Strategy		6/10	Customer intimacy is defensible but harder to prove; the report says so
5. Ethics		5/10	Five concerns; MAC addresses collected that the booking does not require
6. Evidence		8/10	Three of four claims measured; "students prefer it" never was

57 / 80

71.3%

LENS 5

Example 5 — The Code, and Your Assignment Brief

```
# Filling in THIS record is the assessment. Everything else is
presentation.
@dataclass
class SystemUnderReview:
    name: str;          organisation: str
    objectives: List[str] # Lens 1
    people: Dict[str,str] # Lens 2
    process_steps: List[str] # Lens 3
    target_force: str # Lens 4
    ethical_concerns: List[str] # Lens 5
    evidence: Dict[str,str] # Lens 6 - claim -> measurement
    open_questions: List[str]

# The summary finds the LOWEST lens, not the average.
worst_index = scores.index(min(scores))
```

If you cannot fill in the record...

...you have not yet understood the system well enough to assess it. **Discovering that is itself a useful result.**

ASSIGNMENT — due before Week 02

Task. Choose one system you personally use — college portal, delivery app, gym booking, bank app, transit card. Produce a six-lens assessment.

Format. Two pages maximum. One short section per lens. One piece of evidence per lens.

EARNs MARKS: a specific measurement for every claim · at least one thing the system does NOT do well · at least one question you could not answer, and what you would need to answer it.

LOSES MARKS: adjectives without measurements ("very efficient", "user-friendly") · claiming all six business objectives · any lens left blank.

Workflow. Open sheet `05 Six-Lens TOOL`, fill the yellow cells, let it calculate — then write your two pages from what it produces.

What You Have Been Given — All of It Works Offline

`Week01_IS_Digital_Age_LECTURE.md`

The full written lecture — every section on these slides, expanded

`Week01_IS_Digital_Age_SLIDES.pptx`

This deck

`Week01_IS_Reference_Workbook.xlsx`

Glossary · 3 reference tables · 2 LIVE assessment tools · link directory

`code/ — 5 files`

ex1_1 IPO engine · ex1_2 order-to-cash console · ex1_3 five forces · ex1_4 privacy audit · ex1_5 capstone

`diagrams/ — 6 SVG figures`

IPO loop · system types · swimlane · five forces · privacy dataflow · six lenses

`tables/ — 7 CSV files`

Every reference table plus three blank templates, readable in any tool

Nothing requires an account, a licence, or a network connection. SVG files open in any browser. Every required activity can be completed with no internet at all.

Video Protocol, Link Safety, and Legal Notice

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LINK ROT — AND THE FIX

Third-party links break. Videos go private, channels close, platforms reorganise addresses.

Which is why every video entry carries TWO links — a specific video, and a search page. A search page cannot break.

Professor's checklist before each delivery: open the four primary links, confirm each still plays and still runs under fifteen minutes. Two minutes of work.

BEFORE YOU CLICK ANY EXTERNAL LINK

Paste the address into <https://www.virustotal.com/gui/home/url> first.

A link can be correct today and hostile tomorrow — domains expire and get re-registered, pages get compromised, shortened links conceal their destination entirely. Checking takes ten seconds.

AND FOR THE AVOIDANCE OF DOUBT

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END OF WEEK 01 · LESSON 1

Information Systems (IS) in the Digital Age

The one sentence to take away

**An information system is not the computer.
It is the loop — and the person who closes it.**

Before Week 02: re-read sections 1.1.4, 1.3.5 and 1.4.2 · run all five programs at least once · read one professional code of ethics in full and bring one clause you disagree with.

Cyber.SoHo Educational Hub

From the Industry to the Classroom — Building the IT's Future, Today and Together!

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